

DINGTAO, China

WMO: 549090

Lat: 35.0907N

Lon: 115.5232E

Elev: 51

StdP: 100.71

Time Zone: 8.00 (E08)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.8	-6.1	-18.5	0.7	-0.9	-16.0	0.9	-1.7	7.6	2.6	6.9	1.7	1.5	20	0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	34.8	25.0	33.6	24.9	32.4	24.7	28.4	32.2	27.6	31.2	26.9	30.2	3.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.4	23.4	30.9	26.7	22.4	30.2	25.9	21.4	29.2	92.4	32.3	88.9	31.4	85.3	30.3	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	6.2	5.3	-11.5	37.4	1.9	1.4	-12.9	38.4	-14.0	39.3	-15.1	40.0	-16.5	41.1		
			-12.3	29.5	1.8	0.7	-13.7	30.0	-14.7	30.5	-15.8	30.9	-17.1	31.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.8	0.1	3.6	9.6	15.3	20.7	25.8	27.4	26.0	21.6	16.2	8.4	2.1	(6)
(7)		DBStd	9.98	3.10	3.91	4.17	3.83	3.18	2.67	2.40	2.62	2.95	3.46	4.22	3.13	(7)
(8)		HDD10.0	877	307	181	59	5	0	0	0	0	2	78	244		(8)
(9)		HDD18.3	2258	566	412	272	104	13	0	0	7	83	298	503		(9)
(10)		CDD10.0	2626	0	2	46	164	331	474	539	497	348	194	30	0	(10)
(11)		CDD18.3	966	0	0	1	14	86	224	281	239	104	17	0	0	(11)
(12)		CDH23.3	9372	0	0	7	119	731	2352	3054	2257	736	115	2	0	(12)
(13)		CDH26.7	3688	0	0	0	20	222	1070	1315	842	204	15	0	0	(13)
(14)	Wind	WSAvg	2.4	2.4	2.8	3.1	2.9	2.6	2.5	2.2	1.9	1.9	2.1	2.4	2.3	(14)
(15)	Precipitation	PrecAvg	659	5	14	18	31	55	72	174	139	84	31	26	9	(15)
(16)		PrecMax	1082	35	31	74	86	130	203	333	294	295	109	98	36	(16)
(17)		PrecMin	380	0	0	0	5	1	4	51	23	4	0	0	0	(17)
(18)		PrecStd	160	8	10	20	22	33	52	75	85	80	30	28	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	17.2	24.2	29.0	33.9	36.7	36.2	35.0	32.6	28.5	22.4	13.8	(19)
(20)		0.4%	MCWB	5.4	10.7	15.9	20.8	20.3	22.2	26.2	28.1	24.2	17.8	14.4	6.9	(20)
(21)		2%	DB	9.2	14.4	21.6	26.5	31.0	34.9	34.8	33.2	30.6	26.2	19.2	11.0	(21)
(22)		2%	MCWB	3.9	8.2	13.7	18.6	20.7	22.4	26.7	27.3	22.7	17.0	12.3	5.3	(22)
(23)		5%	DB	7.1	12.2	19.2	24.4	29.1	33.6	33.6	32.0	29.0	24.4	17.1	8.9	(23)
(24)		5%	MCWB	2.6	6.6	12.4	17.3	20.3	22.2	26.7	26.5	21.9	16.6	11.2	4.8	(24)
(25)		10%	DB	5.2	10.1	16.9	22.4	27.3	32.2	32.3	30.8	27.5	22.4	15.0	7.2	(25)
(26)		10%	MCWB	1.4	5.6	10.6	16.0	19.6	22.0	26.3	25.9	20.9	15.7	10.0	3.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	11.6	17.4	22.2	23.7	27.0	29.5	29.0	27.1	20.5	15.7	8.4	(27)
(28)		0.4%	MCDWB	10.5	15.9	21.9	27.5	28.6	31.8	33.5	32.9	29.9	24.2	19.8	11.2	(28)
(29)		2%	WB	4.5	9.1	14.9	20.1	22.7	25.6	28.7	28.2	24.4	19.2	13.5	6.7	(29)
(30)		2%	MCDWB	8.2	13.0	19.9	24.9	27.9	30.5	32.7	31.9	27.8	23.3	17.4	9.4	(30)
(31)		5%	WB	3.1	7.5	13.1	18.4	21.6	24.6	27.9	27.6	23.2	18.0	12.2	5.4	(31)
(32)		5%	MCDWB	6.5	11.2	18.3	23.0	26.8	30.0	31.8	30.9	27.0	21.6	15.5	8.2	(32)
(33)		10%	WB	1.8	5.9	11.3	16.9	20.5	23.7	27.2	26.9	22.2	16.9	10.9	4.2	(33)
(34)		10%	MCDWB	4.5	9.4	15.9	21.5	25.5	29.0	30.9	29.7	25.9	20.8	14.2	6.7	(34)
(35)	Mean Daily Temperature Range	MDBR	8.2	8.7	9.7	10.2	10.4	10.4	7.6	7.6	9.2	9.8	9.2	8.4	(35)	
(36)		5% DB	MCDBR	11.2	12.1	12.7	12.5	13.2	12.9	9.5	8.8	11.4	12.7	12.3	11.6	(36)
(37)		5% DB	MCWBR	7.2	7.3	7.1	6.7	5.6	4.0	3.5	3.7	4.4	5.2	6.4	7.2	(37)
(38)		5% WB	MCDBR	9.9	10.4	11.8	11.2	10.6	9.8	8.2	7.6	9.1	9.8	9.9	9.6	(38)
(39)	5% WB	MCWBR	6.6	6.8	7.1	7.0	5.7	4.1	3.8	3.7	4.5	4.8	5.8	6.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.567	0.639	0.644	0.654	0.671	0.805	0.774	0.762	0.708	0.710	0.635	0.567	(40)	
(41)		taud	1.666	1.565	1.580	1.586	1.602	1.415	1.513	1.520	1.571	1.520	1.597	1.691	(41)	
(42)		Ebn at Noon	582	593	646	672	670	584	599	593	594	531	518	547	(42)	
(43)		Edh at Noon	194	240	255	265	264	318	287	280	254	247	207	180	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.23	2.75	4.04	4.79	5.30	5.18	4.61	4.39	3.77	3.09	2.38	2.08	(44)	
(45)		RadStd	0.28	0.45	0.34	0.39	0.40	0.39	0.45	0.57	0.50	0.34	0.35	0.28	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+109	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+89	N/A

Nomenclature: See separate page